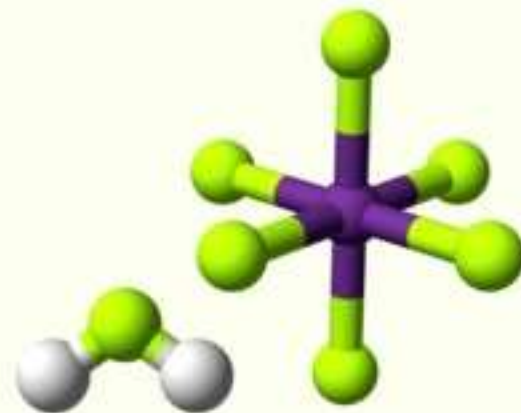


CHEMISTRY

UNDERSTANDING ATOMS

MODULE - 4



HOW HUMANS UNDERSTOOD WHATS INSIDE MATTER ...???

ATOM WAS SUCH A BLACK BOX THAT PROBED THE MINDS OF HUMANS FROM A LONG TIME



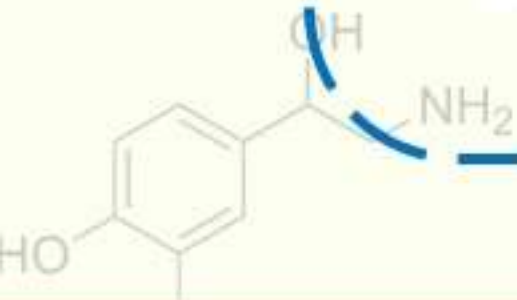
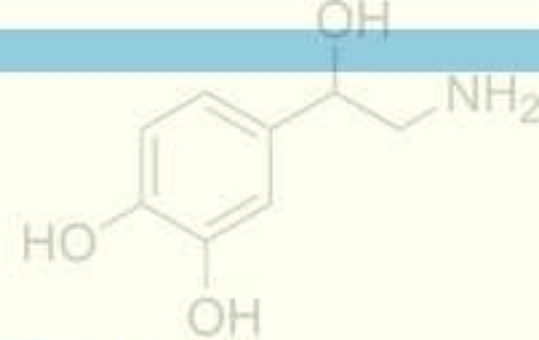
If you have a black box that you cannot open & see how will you know what's inside ?



WHAT WAS THE PRIMITIVE THINKING ABOUT MATTER ...???



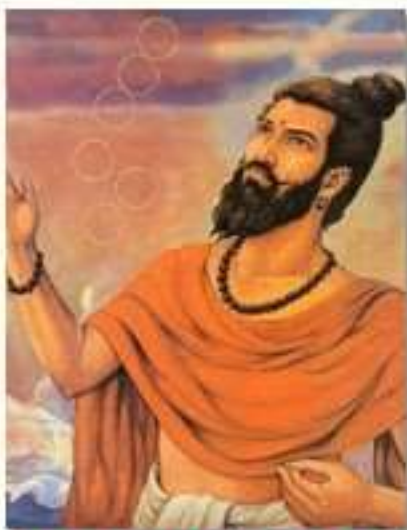
If we continue to divide matter into small and smaller pieces, at one time we will reach the smallest or fundamental unit which will be indivisible !





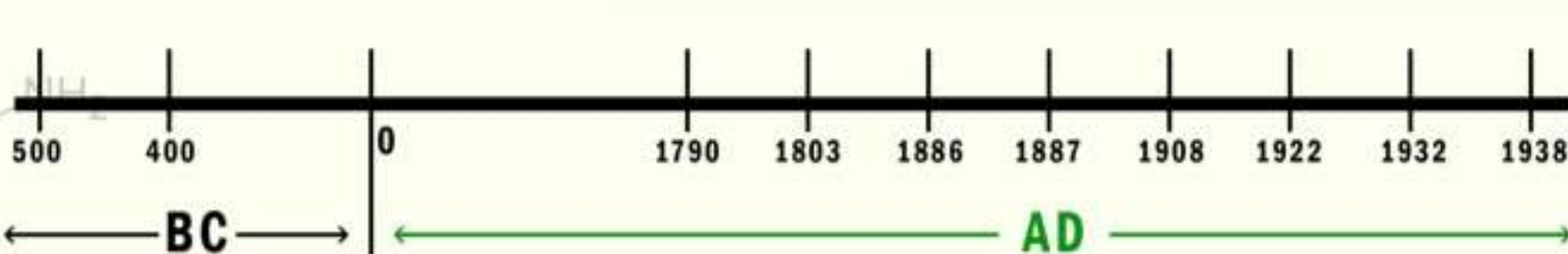
THE SEARCH FOR THE SMALLEST

MAHARISHI KANAD



MAHARISHI KANAD

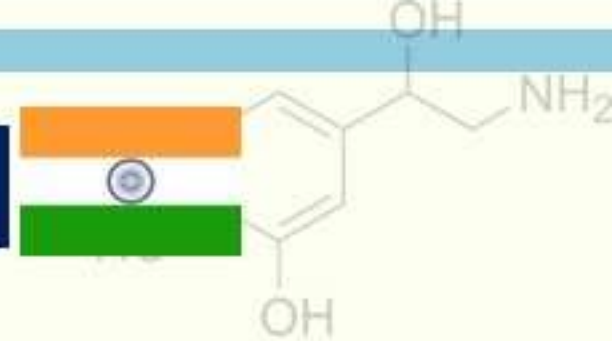
- Around 500 BC, Maharishi Kanad propounded atomic theory.
- He postulated that if '*padarth*' or matter can be divided into smaller and smaller particles, a time will come when no further division is possible.
- He named this smallest particles of a *padarth* as '*Parmanu*'.



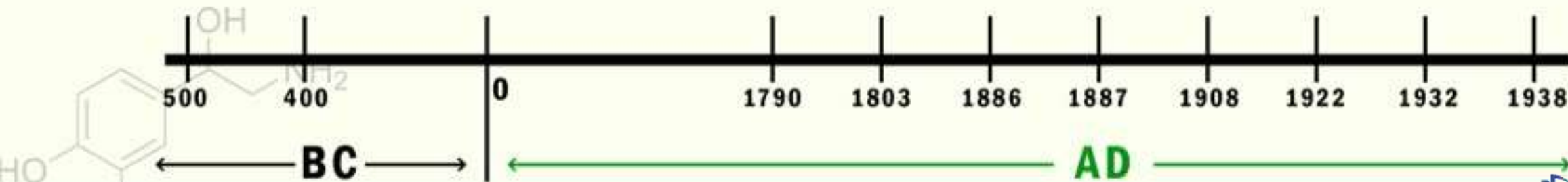


THE SEARCH FOR THE SMALLEST

PAKUDHA KATYAYAMA



- Pakudha Katyayama elaborated the doctrine of Maharishi Kanad.
- He said that these particles exist in combined form.
- The particles form various forms of matter.





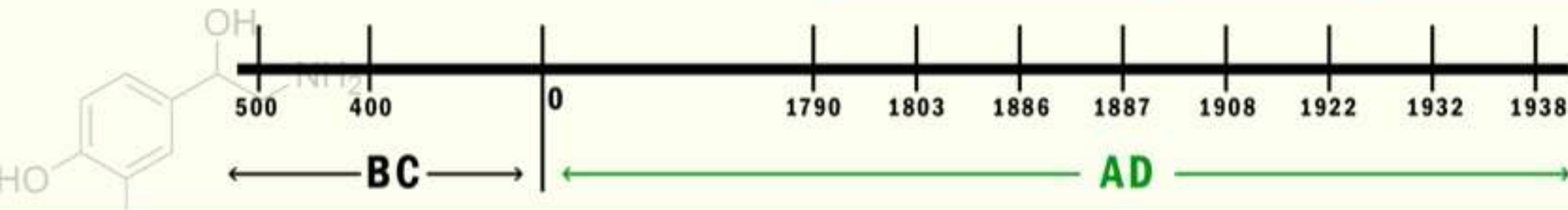
THE SEARCH FOR THE SMALLEST

DEMOCRITUS AND LEUCIPPUS



**DEMOCRITUS &
LEUCIPPUS**

- Two Greek philosophers around the same era gave their theory on the smallest particle.
- They suggested that if we go on dividing matter, it will reach its indivisible state.
- Democritus called this Atoms or 'the indivisible'.
- These were philosophical considerations but were not experimentally verified.





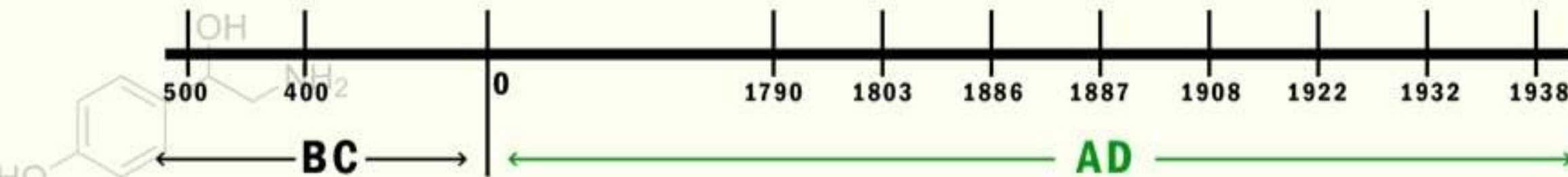
THE SEARCH FOR THE SMALLEST

ANTOINE LAVOISIER



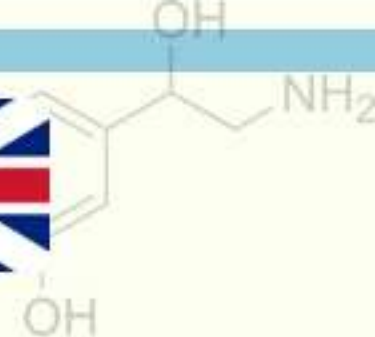
ANTOINE LAVOISIER

- In the 18th century Antoine Lavoisier laid the foundation of chemical Sciences.
- He established two important laws of chemical combination.
- He was the first of his time to research on various elements.





THE SEARCH FOR THE SMALLEST

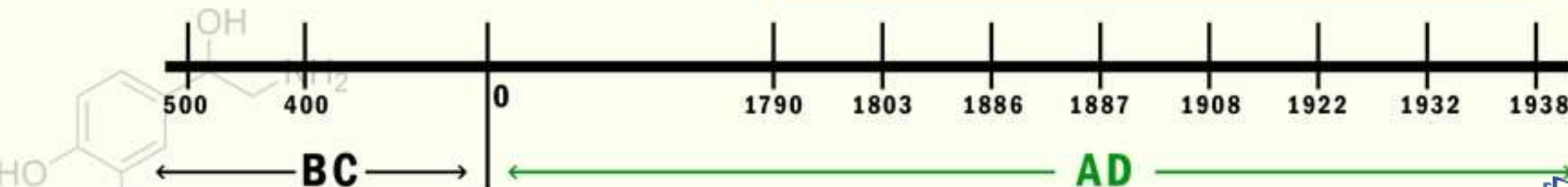


JOHN DALTON



JOHN DALTON

- In the year 1803 Dalton proposed the Atomic Theory.
- He postulated that atom cannot be created, destroyed or subdivided.
- Every different elements has different atoms(size, mass).
- In chemical reactions atoms were combined separated and rearranged.





THE DISCOVERY OF SUBATOMIC PARTICLES



PROTON



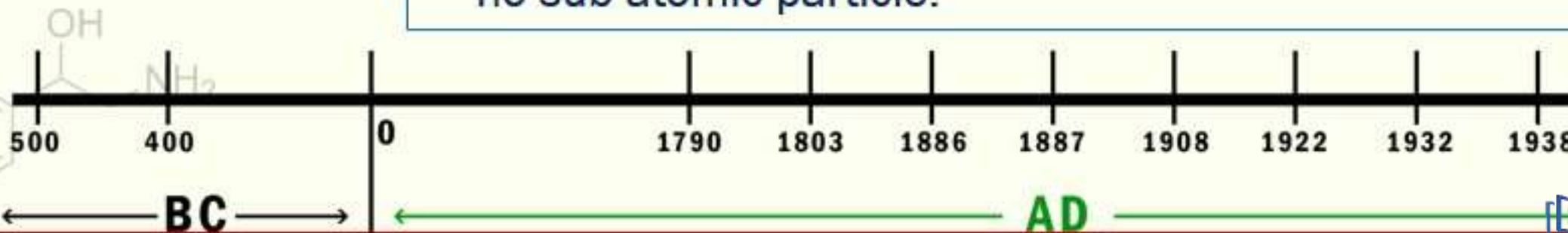
**EUGEN
GOLDSTEIN**



CANAL RAYS

- E. Goldstein in 1886 discovered positively charged radiations.
- He named them '*Canal Rays*'

- It was positively charged and 2000 times more heavier than electron.
- It was named proton, p^+ , whose charge is + and mass is 1 unit.
- John Dalton was proved wrong as he said that there is no sub atomic particle.





THE DISCOVERY OF SUBATOMIC PARTICLES



ELECTRON

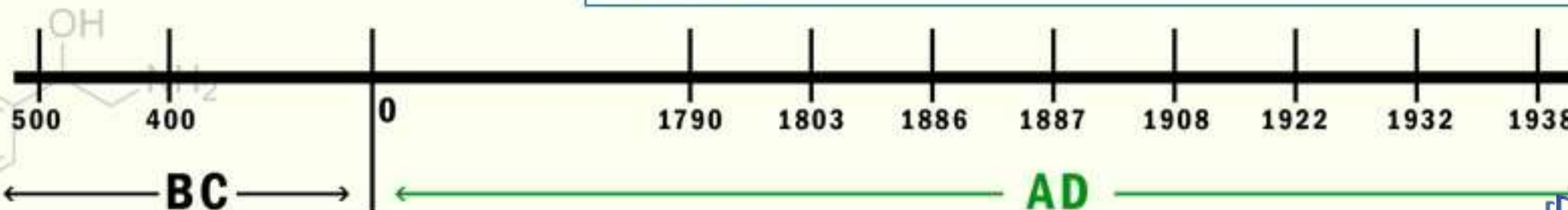


JOSEPH
JOHN
THOMPSON



JOHNSON
STONEY

- In 1887, Thompson was studying Electric discharges in Cambridge England.
- He observed low pressure gases glows in high voltage.
- These rays came from cathode - the negative end.
- In 1891, Johnson Stoney named it Electron.
- Thomson tried to put forth a atomic model as well.



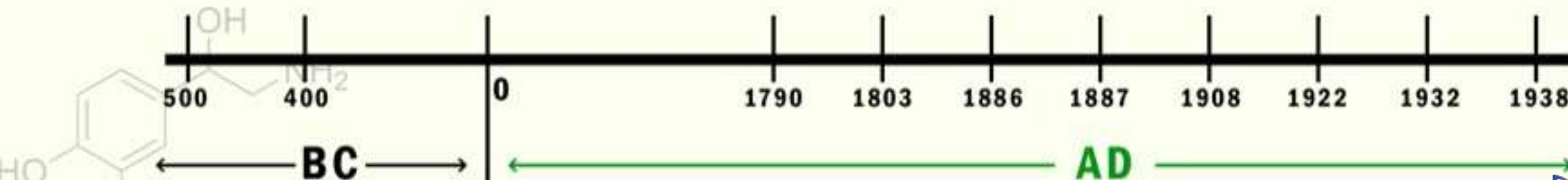
THE DISCOVERY OF SUBATOMIC PARTICLES



THE CHRISTMAS PUDDING MODEL - THOMPSON



- Thomson proposed the model of an atom to that of a Christmas pudding.
- He proposed that the sphere is positively charged.
- The negative electrons are embedded in it.



THE SEARCH FOR THE ATOMIC MODEL

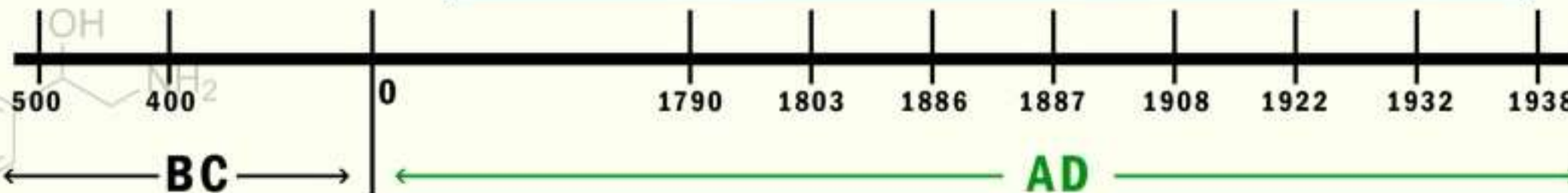


ERNEST RUTHERFORD



ERNEST RUTHERFORD

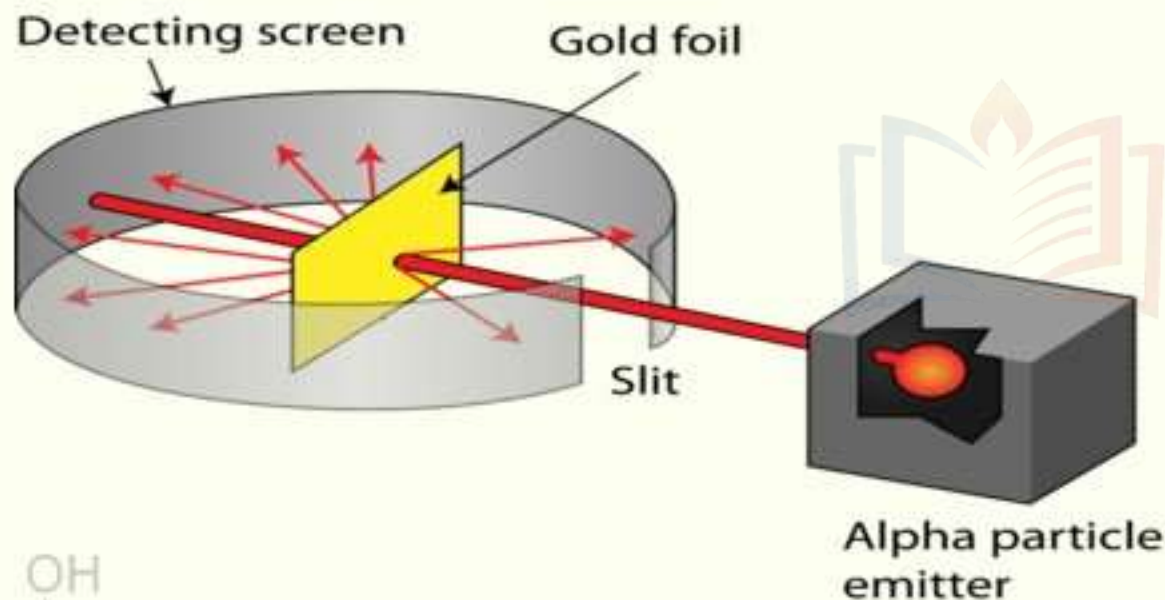
- Rutherford was interested in knowing how the electrons are arranged within the atom.
- He designed an experiment to study the atom.
- This experiment came to be known as the famous **Gold Foil Experiment**.
- He charged a gold foil with high energy α -particles.
- α -particles are produced from heavy metals like uranium by radioactive decay.



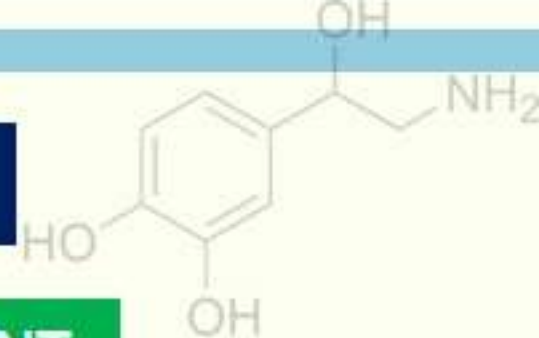
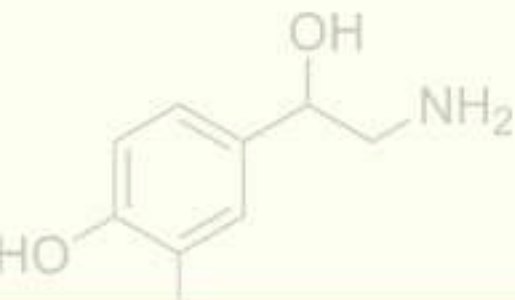


THE SEARCH FOR THE ATOMIC MODEL

THE GOLD FOIL α -PARTICLE SCATTERING EXPERIMENT



- The gold foil was only 1000 atoms thick.
- He placed the gold foil inside a fluorescent detecting screen.
- α -particles would hit the foil and the emitted rays would be detected by fluorescence on the screen.

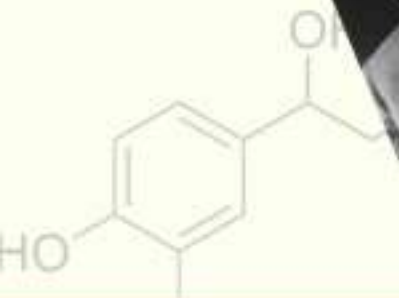




ANOMALY BETWEEN EXPECTED AND OBSERVED RESULTS...



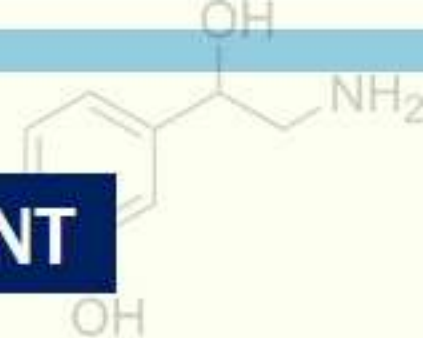
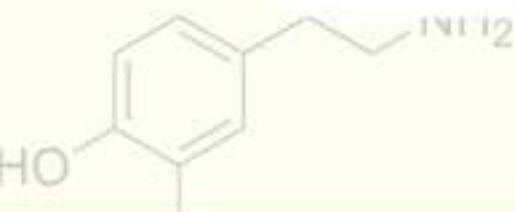
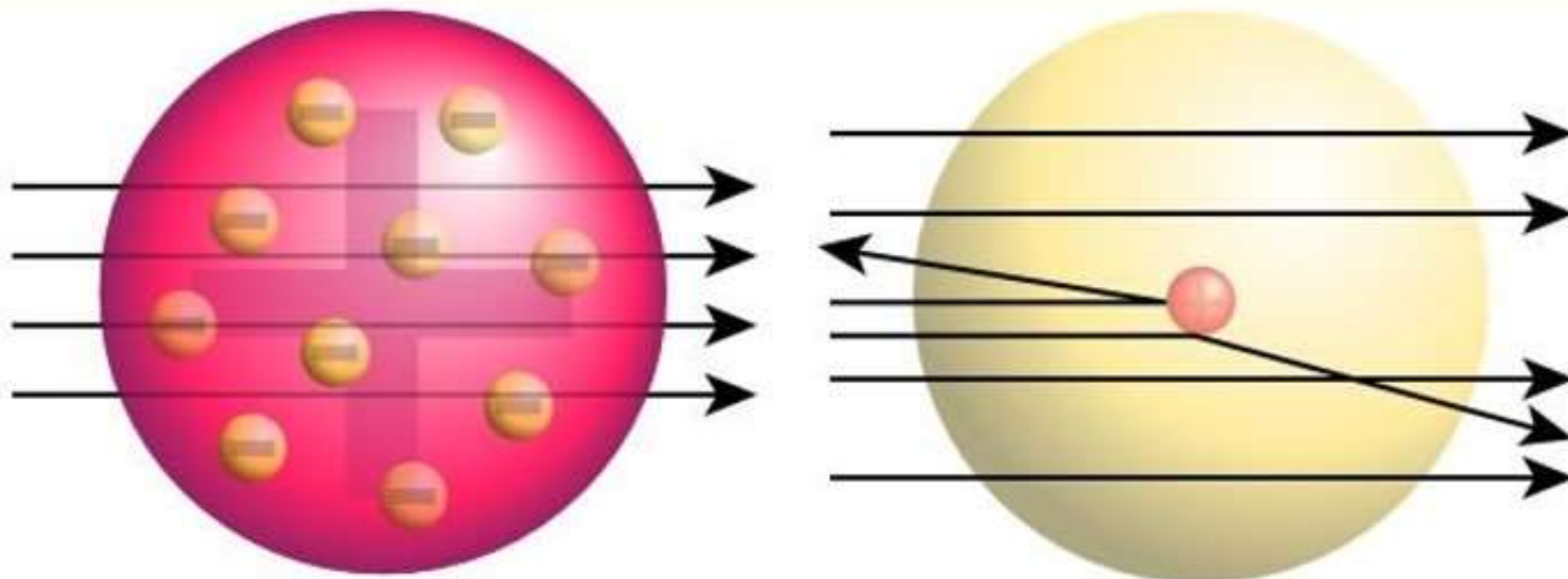
*Rutherford felt as if you hit
a 15 inch shell on a tissue
paper and it comes back
and hit you!!!*





THE GOLD FOIL α -PARTICLE SCATTERING EXPERIMENT

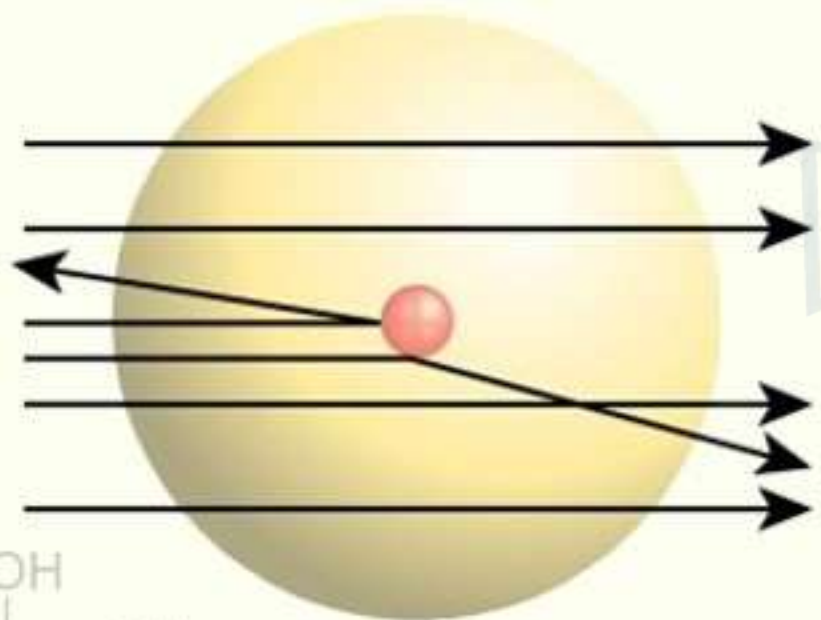
EXPECTED vs. OBSERVED RESULTS!



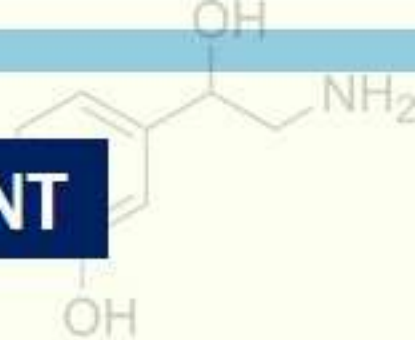
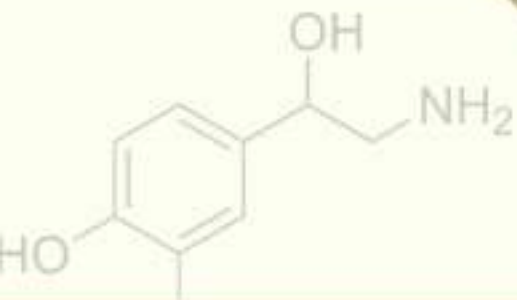


THE GOLD FOIL α -PARTICLE SCATTERING EXPERIMENT

OBSERVED RESULTS!

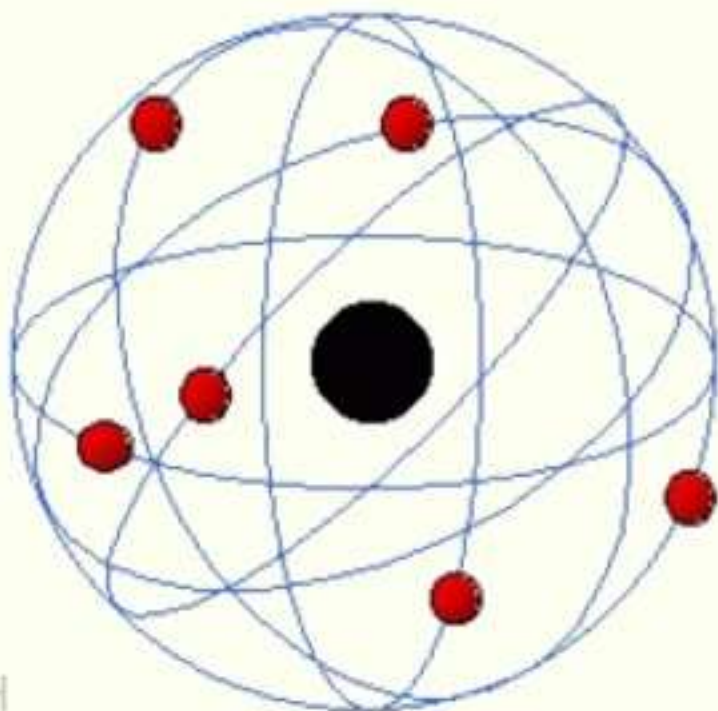


- Most of the alpha particles passed straight through the foil.
- Some α -particles showed deflection by small angles.
- One out of 12,000 particles surprisingly rebounded which is not expected in Thomson's model.
- This could only happen if the α -particles have hit a concentrated positive charge.





RUTHERFORD'S MODEL OF AN ATOM



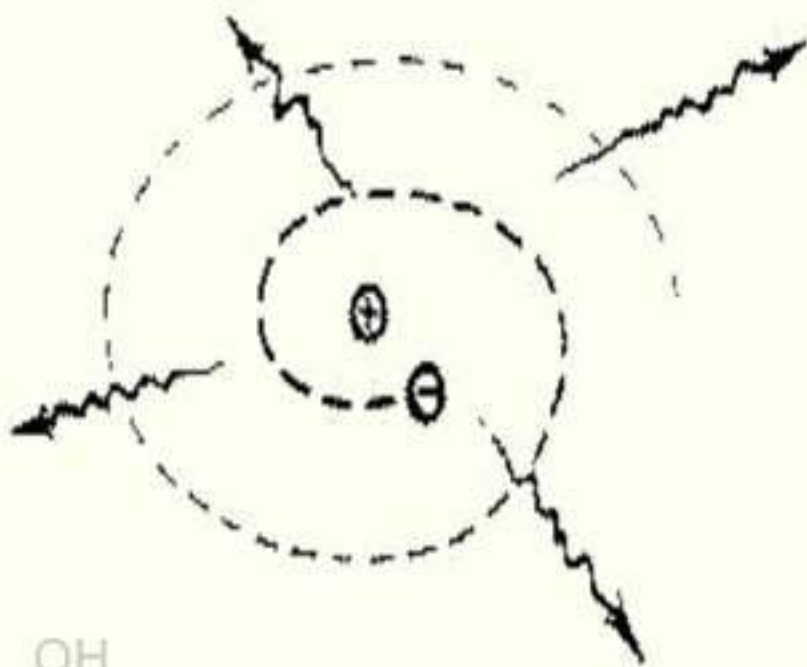
- The atom is an positively charged centre called nucleus.
- Nearly all the mass of the atom is concentrated in the nucleus.
- The electron revolve around the nucleus in circular orbits.
- The size of the nucleus is very small compared to the atom.
- A tennis ball to a large stadium is a nucleus to the atom.





BUT WHY DID RUTHERFORD'S MODEL FAIL?

ELECTRONS CANNOT REVOLVE AROUND ANY ORBIT !



- Electrons revolving in any orbit is not stable.
- Electrons will accelerate when revolving as they are charged particle.
- During acceleration, electrons will radiate energy.
- Thus, they will lose energy and collapse into the nucleus.
- Hence matter will no longer exist.

Rutherford couldn't explain this anomaly thus, his model was not accepted



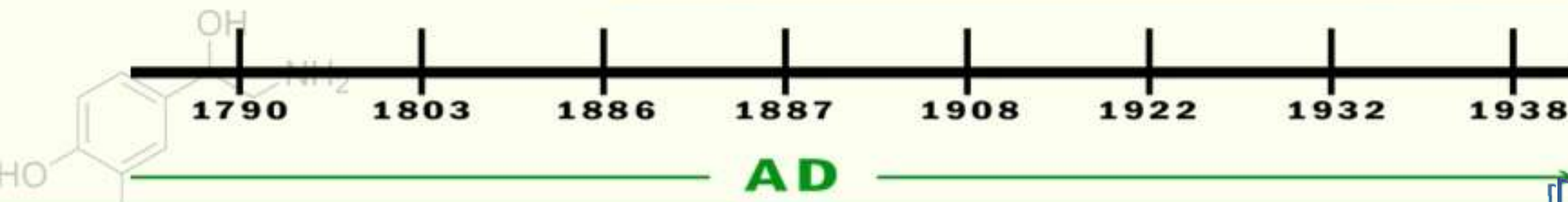


THE MODIFICATION TO THE ATOMIC MODEL – NEILS BOHR



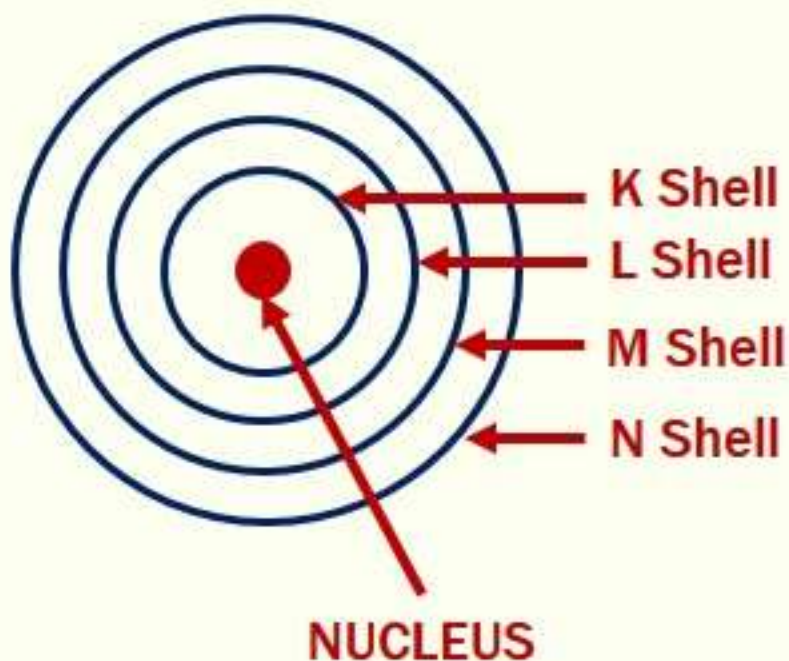
NEILS BOHR

- A Danish physicist, Neils Bohr adapted Rutherford's nuclear structure and created Bohr's atomic model .
- His atomic model included Hydrogen and Hydrogen like atoms like Helium.
- He also published books like - The Theory Of Spectra, Atomic Theory and Description of Nature.
- He got Nobel Prize for atomic structure in 1922.





THE BOHR'S ATOMIC MODEL



- 1 Electrons revolve in discrete orbits.
- 2 Electrons do not radiate energy while revolving in the **discrete orbits**.
- 3 These orbits or shells are called energy levels. They can be named K,L,M,N or numbered 1,2,3,4
- 4 Electrons radiate an specific amount of energy when they move from higher to lower orbits. Electrons absorb specific amount of energy when they climb to higher orbits.





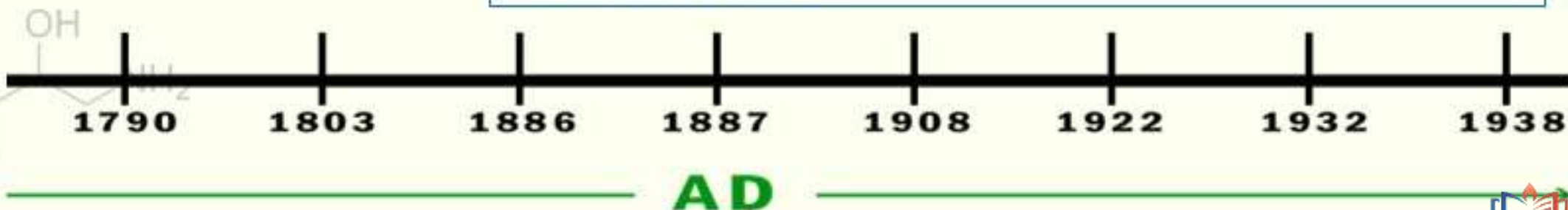
WHAT MISSED IN THE PREVIOUS MODELS...???

NEUTRON



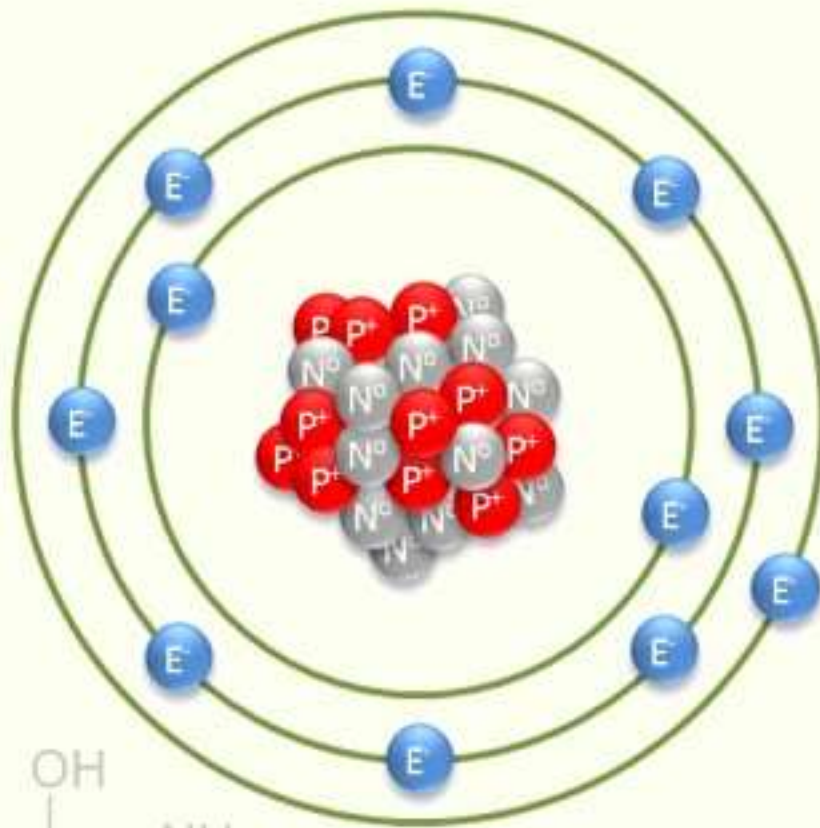
JAMES CHADWICK

- In **1932**, J. Chadwick discovered another subatomic particle.
- The particle had no charge but mass like proton.
- It was named as Neutron.
- Neutrons are present in all atoms **but Hydrogen**.
- It also contributes to the atomic mass and is another member of the nucleons.
- Chadwick was awarded Nobel Prize in 1935 for his discovery.

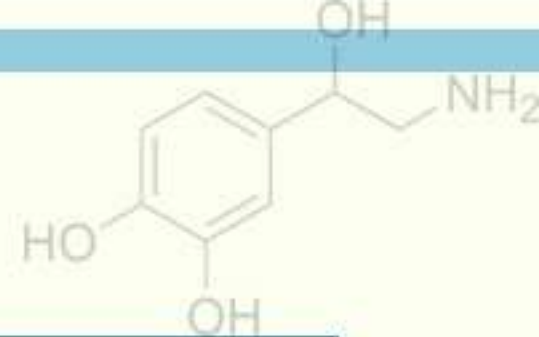
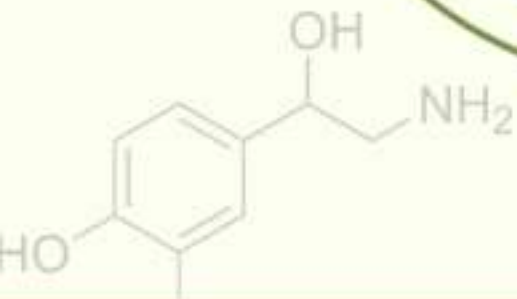




THE FINAL STRUCTURE OF ATOM



- An atom is a stable elementary unit of matter.
- An atom consists of proton, neutron and electron.
- Proton and neutron are called nucleons.
- Electrons are attracted towards the proton by electrostatic force.
- The number of protons and electrons are equal.
- The number of protons and neutrons may not be equal.



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